

Power To The PC

Faulty power supply can spell doom for PCs. Agent 001 tracks down the better ones

A month ago our editor SK, decommissioned his five-year-old glorified typewriter—his older Pentium pro system—with the idea of getting a new 64-bit super fast desktop PC. While he is not a geek you often come across in the Test Centre (TC) or *Skoar!*, his six-month stay amongst them has made him tech savvy enough to long for a 64-bit system. So the TC drew up the best configuration and I gave my inputs for the best market deals, all set and done the final piece of paper was given to a local assembler near SK's house. He assembled the beauty, and boy, oh boy the smile on SK's face was infectious when the monster of a machine zoomed through most applications like a Ferrari on an open road.

Everything was wow, cool and fast until that fateful Sunday morning when a high pitch started emanating from the box and soon, everything went cold. The TC was summoned, and they traced the fault to a fishy power supply (PS). But why wasn't it a 400W supply with dual fans? Just the question to put a sleuth like me on the right track.

This time around, though, the job was difficult since no one in the TC could come up with quantitative proof as to which power supply brand is more reliable or efficient. However, from their testing experiences, they have reservations about certain brands known to be better than others. According to them, a 350W PS is a must if you own one of those deluxe, platinum boards overflowing with additional features and demanding more power.

But is putting a higher capacity PS the only solution? Debating on the issue, we arrived at the conclusion that a better (read efficient) 350W PS is as good as say, a poor 450W PS.

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Efficiency in this case refers to the ability of the PS to fulfil the demand from the machine without significant internal loss. For example, if a PS is rated at 350W, under typical load conditions, the total power consumed was 250W whereas the power supplied to the system was 175W. This amounts to 70 per cent efficiency implying that the remaining 30 per cent was internal loss and dissipated in the form of heat. And all this under typical load conditions. The consumption would vary depending on the work done by the processor.

Why processors? Processors are the real power hungry monsters in the machine. At low loads, when the system demands less power, the amount of energy lost will be high presenting a precarious situation for a poorly designed high-wattage PS since most of the power consumed will be flushed out as heat rather than used by the machine. Therefore, although a good PS is expensive, in the longer run, it will prove to be cost effective.

So which one should you buy? Visit almost any computer shop and a 300W PS costing Rs 500 will be the first one handed out to you. Brands are dime-a-dozen in this category and you would be forgiven for thinking that just about everybody in the world has a manufacturing plant! Not all 300W PS, though, are bad. In fact, Mercury, Microtek, Frontech, Perx and HIS, to name a few, are the fastest moving ones and from my investigation, they don't fare too badly either. However, when confronted with the efficiency question, most shopkeepers will be at a loss to answer you, well, efficiently!

Regular 400W PS are a different ballgame altogether. Everybody knows the ratings are bogus and stickers are just that—stickers. Ask for a 400W PS and be ready to spend up to Rs 1.5K for a bogus PS. Push a little harder and reluctantly, you might hear the name VIP. The 400W PS from VIP is priced at Rs 1.9K and comes with a three-year warranty and according to shopkeepers is the best you can buy. Our TC swears by Antec, a name not familiar to most in the market. They are priced at a premium but are worth the price.

In the marketplace, VIP attracts attention as being the best amongst the brands. Antec is definitely better and has a wide product portfolio to choose from but is priced higher. With no proper test in place, recommending a buy is difficult but I will bet my money on an Antec or a VIP.

Want more of Agent 001? Turn over to read his **Answers** to your buying questions



Illustration: Shyam Shiram